

PHY201 : Waves & Optics

Second Mid-Semester Examination 2025 — Detailed Solutions

1. (a) Let $x_1(t), x_2(t)$ be the small horizontal displacements of the two bobs (so $x = L\theta$ for small angles). An isolated pendulum has small-angle frequency $\omega_0 = \sqrt{g/L}$. With a coupling spring of constant k between the bobs, the linearized equations are

$$m\ddot{x}_1 + m\omega_0^2 x_1 + k(x_1 - x_2) = 0, \quad m\ddot{x}_2 + m\omega_0^2 x_2 + k(x_2 - x_1) = 0.$$

Seek normal modes $x_j(t) = A_j e^{i\omega t}$. Then

$$\begin{bmatrix} m(\omega_0^2 - \omega^2) + k & -k \\ -k & m(\omega_0^2 - \omega^2) + k \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \mathbf{0},$$

and nontrivial amplitudes require

$$(m(\omega_0^2 - \omega^2) + k)^2 - k^2 = 0 \implies m(\omega_0^2 - \omega^2) + k = \pm k.$$

Hence the two eigenfrequencies are

$$\omega_+ = \omega_0 \quad (\text{in-phase : } A_1 = A_2), \quad \omega_- = \sqrt{\omega_0^2 + \frac{2k}{m}} \quad (\text{out-of-phase : } A_1 = -A_2).$$

(b) If one bob is displaced and released, the initial state is a superposition of the two normal modes. Because $\omega_- \neq \omega_+$, their relative phase drifts at rate $\Delta\omega = \omega_- - \omega_+$, producing a slow beat envelope: energy oscillates between the two pendulums, being mainly in one while the other is nearly at rest, and then swapping after half a beat.

(c) With $L = 0.5$ m and $g \simeq 10$ m/s²,

$$\omega_0 = \sqrt{\frac{g}{L}} = \sqrt{20} = 4.472 \text{ s}^{-1} \quad (\text{given}).$$

A 2% upward shift means $\omega_- = 1.02 \omega_0$, so

$$\Delta\omega = \omega_- - \omega_+ = 0.02 \omega_0 = 0.02 \times 4.472 \approx 0.08944 \text{ s}^{-1}.$$

Therefore the beat (envelope) period is

$$T_{\text{beat}} = \frac{2\pi}{\Delta\omega} \approx \frac{2\pi}{0.08944} \approx 7.025 \times 10^1 \text{ s} \approx 70.3 \text{ s}.$$

2. (a) Let $y(x, t)$ denote the small transverse displacement of a taut string of length L , linear density μ , under tension T . It satisfies

$$y_{tt} = v^2 y_{xx}, \quad v = \sqrt{\frac{T}{\mu}}.$$

Fixed ends at $x = 0$ and $x = L$ impose

$$y(0, t) = 0, \quad y(L, t) = 0 \quad \text{for all } t.$$

Seek separable standing waves $y(x, t) = X(x)T(t)$. Substituting,

$$\frac{T''(t)}{T(t)} = v^2 \frac{X''(x)}{X(x)} = -\omega^2,$$

so

$$X'' + k^2 X = 0, \quad T'' + \omega^2 T = 0, \quad \omega = vk.$$

Impose $X(0) = 0$ and $X(L) = 0$. The nontrivial solutions of $X'' + k^2 X = 0$ satisfying both boundary conditions are

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right), \quad k_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots$$

Hence the allowed angular frequencies are

$$\omega_n = v k_n = \frac{n\pi v}{L},$$

and the (ordinary) frequencies are

$$f_n = \frac{\omega_n}{2\pi} = \frac{nv}{2L} = \frac{n}{2L} \sqrt{\frac{T}{\mu}}, \quad n = 1, 2, 3, \dots$$

with mode shapes $y_n(x, t) = A_n \sin\left(\frac{n\pi x}{L}\right) \cos(\omega_n t + \phi_n)$.

(b) Wave speed:

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{100}{10^{-2}}} = \sqrt{10^4} = 100 \text{ m/s}.$$

Therefore

$$f_n = \frac{nv}{2L} = \frac{n \cdot 100}{4} = 25n \text{ Hz},$$

so

$$f_1 = 25 \text{ Hz}, \quad f_2 = 50 \text{ Hz}, \quad f_3 = 75 \text{ Hz}.$$

(c) Since 78 Hz is closest to $f_3 = 75 \text{ Hz}$ (and farther from $f_4 = 100 \text{ Hz}$ and $f_2 = 50 \text{ Hz}$), the **third mode** ($n = 3$) will resonate most strongly:

$n = 3$ resonates most strongly, as the drive is nearest to f_3 .

3. (a) Consider a small longitudinal element of the rod. Let $\xi(x, t)$ be the longitudinal displacement, ρ the mass density, and Y Young's modulus. For a linear, lossless rod the longitudinal wave equation is

$$\xi_{tt} = c^2 \xi_{xx}, \quad c = \sqrt{\frac{Y}{\rho}}.$$

Stress and strain are related by Hooke's law:

$$\sigma(x, t) = Y \varepsilon(x, t) = Y \frac{\partial \xi}{\partial x}.$$

The *kinetic energy density* (energy per unit volume) is

$$u_K = \frac{1}{2} \rho \left(\frac{\partial \xi}{\partial t} \right)^2.$$

The *elastic potential energy density* equals (work per unit volume stored in strain)

$$u_P = \frac{1}{2} \sigma \varepsilon = \frac{1}{2} Y \left(\frac{\partial \xi}{\partial x} \right)^2.$$

For the travelling wave $\xi(x, t) = \xi_0 \cos(kx - \omega t)$,

$$\xi_t = \omega \xi_0 \sin(kx - \omega t), \quad \xi_x = -k \xi_0 \sin(kx - \omega t),$$

hence

$$u_K = \frac{1}{2} \rho \omega^2 \xi_0^2 \sin^2(kx - \omega t), \quad u_P = \frac{1}{2} Y k^2 \xi_0^2 \sin^2(kx - \omega t).$$

Using the dispersion $\omega = ck$ and $Y = \rho c^2$, we have $Y k^2 = \rho \omega^2$, so

$$u_P = u_K, \quad u = u_K + u_P = \rho \omega^2 \xi_0^2 \sin^2(kx - \omega t),$$

and the time averages are

$$\langle u_K \rangle = \langle u_P \rangle = \frac{1}{4} \rho \omega^2 \xi_0^2, \quad \langle u \rangle = \frac{1}{2} \rho \omega^2 \xi_0^2.$$

(b) The instantaneous power crossing a unit area (energy flux) equals stress $\times$ particle velocity:

$$S = \sigma \xi_t = Y \xi_x \xi_t.$$

For the given wave,

$$S = Y(-k \xi_0 \sin \phi)(\omega \xi_0 \sin \phi) = Y k \omega \xi_0^2 \sin^2 \phi, \quad \phi \equiv kx - \omega t.$$

Thus the time-averaged flux is

$$\langle S \rangle = \frac{1}{2} Y k \omega \xi_0^2 = \frac{1}{2} (\rho c^2) \left(\frac{\omega}{c} \right) \omega \xi_0^2 = \frac{1}{2} \rho c \omega^2 \xi_0^2 \propto \xi_0^2 \omega^2 c.$$

So the average energy flux is proportional to $\xi_0^2 \omega^2 c$ with proportionality constant $\frac{1}{2} \rho$.

4. (a) Given (in cm): $y_1 = 0.50 \cos(20x - 100t)$, $y_2 = 0.50 \cos(20x - 104t)$. Using $\cos a + \cos b = 2 \cos \frac{a+b}{2} \cos \frac{a-b}{2}$,

$$\begin{aligned} y &= y_1 + y_2 = 2 \cdot 0.50 \cos(20x - 102t) \cos(2t) \\ &= \cos(20x - 102t) \cos(2t) \quad (\text{cm}). \end{aligned}$$

(b) Carrier frequency $\omega_c = 102 \text{ s}^{-1}$, beat frequency $\omega_b = 2 \text{ s}^{-1}$, so $f_{\text{beat}} = \omega_b / 2\pi \approx 0.318 \text{ Hz}$. Physically, the amplitude envelope oscillates at f_{beat} due to the interference of close frequencies.

(c) Maximum transverse speed occurs when $|\dot{y}|$ is maximal. For a single sinusoid $A \cos(\omega t)$, $|v_{\text{max}}| = A\omega$. Here the instantaneous amplitude is modulated; the peak particle speed occurs near envelope maxima with carrier $\omega_c = 102 \text{ s}^{-1}$ and amplitude $A_{\text{max}} = 1.0 \text{ cm}$. Hence

$$v_{\text{max}} \approx A_{\text{max}} \omega_c = (1.0 \text{ cm}) \times 102 \text{ s}^{-1} = 102 \text{ cm/s} \quad (= 1.02 \text{ m/s}).$$